

Rajat Dhanraj Savdekar

Los Angeles, CA | (213) 574-8894 | rsavdeka@usc.edu | LinkedIn | github.com/RDX-Rajat-Savdekar | rajatsavdekar.dev

EDUCATION

University of Southern California

Masters of Science in Computer Science

Relevant Coursework: Game Design, Multimedia System, Cyber Physical Systems

Los Angeles, California

Aug 2024 – **May 2026**

GPA: 3.6

University of Mumbai

Bachelor of Engineering in Computer Engineering

Relevant Coursework: Computer Networks, Software Engineering, Data Structures, AR/VR

Mumbai, India

Aug 2019 – May 2023

GPA: 3.7

SKILLS

Languages: Python, Java, JavaScript, C++, SQL

Frontend: React, Next.js, Angular, HTML/CSS

AI/ML: PyTorch, LLMs, RAG, Scikit-learn

Databases: PostgreSQL, MySQL, MongoDB, Vector DBs

Backend: Node.js, Flask, REST APIs, Microservices

Cloud & DevOps: AWS, GCP, Azure, Docker, CI/CD, Git

WORK EXPERIENCE

Software Engineer

Jalgaon Fruits Sales Cooperative

May 2023 – May 2024

Jalgaon, India

- Architected and launched a full-stack procurement and inventory management system for **150+ daily users**, digitizing manual workflows to reduce data entry errors by over 90%.
- Refactored a monolithic PHP application into scalable **Flask microservices**, improving system reliability and reducing API response times during peak usage.
- Implemented a complete DevOps lifecycle with **Docker and CI/CD pipelines (GitHub Actions)**, accelerating release cycles by ~30% and ensuring 99.9% uptime during harvest seasons.
- Developed real-time analytics dashboards using **React and Flask**, providing data-driven insights that cut produce spoilage by 15% through better inventory tracking.

Teaching Assistant

USC Summer Programs

May 2025 – July 2025

Los Angeles, California

- Co-designed and led hands-on C++ and circuit design workshops in partnership with SoCalGas for over 160 students.
- Mentored student teams on sustainability-focused engineering projects, resulting in the completion of 20+ unique, student-led prototypes.

PROJECTS

Aura: Live Caption & Sound Awareness

[\[Youtube\]](#) [\[GitHub\]](#)

- Developed a prize-winning visionOS app to enhance awareness for the hard of hearing using AR alerts.
- Implemented live speech transcription in a **RealityKit** immersive space using **SwiftUI** and the **Speech** framework.
- Integrated a **CoreML** model for on-device sound classification (e.g., sirens), providing users with critical visual alerts.

LeetCode Study Hub

[\[Live\]](#) [\[GitHub\]](#)

- Engineered a full-stack MERN application to enhance long-term knowledge retention for DSA concepts by combining a problem tracker with an intelligent flashcard system based on a spaced repetition algorithm.
- Developed a secure **Node.js** REST API with **JWT** authentication for all user and problem data management.
- Integrated the **Google Gemini API** to create an AI assistant that suggests algorithmic patterns from problem text.

SplitIt - A Splitwise Clone

[\[Live\]](#) [\[GitHub\]](#)

- Engineered a full-stack Splitwise clone with a secure **Node.js** API and a dynamic **React** UI using **Tailwind CSS**.
- Implemented a complex backend algorithm to simplify transaction data into a clear, real-time balance sheet for users.

DSA and System Design Concepts Videos

[\[GitHub\]](#)

- Created 20+ animated visualizations of complex DSA and system design concepts using Python and Manim.
- Engineered a modular rendering pipeline with caching, reducing average visualization build time by 30%.

Image Texture Analysis with GLCM

[\[Publication\]](#)

- Authored a highly-cited research paper (**10+ citations** in IEEE, Springer, etc.) on a novel image texture analysis tool.
- Developed a MATLAB GUI to compute key texture features, providing a practical analysis tool for researchers.

COMMUNITY LEADERSHIP & MENTORSHIP

- Mentorship & Tech Outreach:** Supported USC Viterbi Impact Group by mentoring students in technical challenges (**Hustle N Code**, **SoLA Robotics**, **MESA K12**).
- Global Volunteerism:** Contributed to community initiatives, including food distribution (**LA Food Bank**) and animal rescue drives (**Pashu Papa**, **NGO**).